

Spatially-Correlated Distributionally Robust Scheduling for Carbon-Aware Data Centers

Phase 1 Results — The Spatial Question, Resolved

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Abstract

This chapter closes the central Phase 1 question: does modelling the *spatial correlation* of carbon intensity across electrically coupled regions improve a distributionally robust (DRO) data-center schedule? Extending the methodology of the status report to **three** US/Canada cases that span the full correlation spectrum, we find a **replicated, robustness-checked null**: the joint spatial covariance never beats an independent (shuffled-marginals) baseline by a material margin. The result is neither an artifact of weak correlation — we first *validate* strong, de-seasonalized correlation in two interconnections — nor an assertion: a **mean-ablation experiment** shows the covariance signal is genuine and exploitable but *dominated* by the mean carbon field, and a **tail-dependence** diagnostic shows the residual structure is non-elliptical (upper-tail-independent, radially asymmetric) and therefore invisible to a covariance-based ambiguity set by construction. Phase 1 thus delivers a *specification result* that directly motivates the Phase 2 copula extension.

1 Headline

Spatial correlation of carbon intensity does not translate into robust scheduling value: across the full correlation spectrum, in a Mahalanobis–Wasserstein DRO, modelling the joint spatial distribution buys essentially nothing over treating each region’s carbon independently.

Three findings make this a substantive, defensible result rather than a bare negative:

1. **It is not absence of correlation.** We validate strong, residual-surviving correlation in two interconnections (§??); the DRO’s spatial assumption *is valid*. Validity of the assumption $\neq$ value from it.
2. **It is not an edge case.** The null holds in the strongly-correlated cases, in an adversarial *heterogeneous* case engineered to be diversifiable, and through a full robustness battery with multiple-testing correction (§??–§??).
3. **We prove *why*, causally.** A mean-ablation experiment (§??) demonstrates the covariance is real but mean-dominated; a tail-dependence diagnostic (§??) shows the exploitable structure is non-elliptical.

2 The three cases: a correlation spectrum

We transport the *identical* machinery of the status report — Algorithm 2b Mahalanobis–Wasserstein second-order cone program (SOCP), three nested constraint regimes, CVaR_{0.95} metric, blocked 5-fold cross-validation to select the Wasserstein radius ε , and a 1000-resample bootstrap confidence interval on the spatial gap — to three new region sets chosen to span the dependence spectrum (Table ??). All panels are hourly Electricity Maps lifecycle carbon intensity, 2021–2025 (43,824 hourly observations per zone).

Case	Zones	Residual correlation	Mechanism
<code>us_west</code>	CISO, BANC, LDWP, NEVP, AZPS	strong (0.72–0.78)	WECC: solar + weather
<code>taskc</code>	CA-ON, NYIS, MISO, PJM	strong (0.70–0.73)	East: weather fronts
<code>us_hetero</code>	CISO, ERCO, BPAT	near-zero (ERCO–BPAT 0.00)	solar / wind / hydro

Table 1: The three Phase 1 cases. `us_hetero` is adversarial by design: regions with different generation physics, chosen so their clean/dirty periods do *not* coincide — if spatial DRO value exists anywhere, it should appear here.

Correlation is genuine and survives de-seasonalization. Removing the per-zone hour-of-day mean (which collapses the spurious diurnal co-movement seen in the California and Iberia sub-grids of the status report), the two interconnected cases retain strong correlation: `us_west` CISO–NEVP 0.78, CISO–LDWP 0.72; `taskc` CA-ON–NYIS 0.73, MISO–PJM 0.70. This confirms the hypothesis that interconnected US/Canada grids carry real spatial carbon structure, unlike the European cases. Figure ?? (and `figures/ci_corr_heatmap_*`) shows the full correlation matrices.

3 The spatial gap: no robust value

The falsification test pairs the scheduler with either the *joint* sample covariance $\hat{\Sigma}$ or its *block-diagonal* (cross-region-zeroed) counterpart $\hat{\Sigma}^{\text{shuf}}$; the out-of-sample CVaR_{0.95} gap shuf – joint measures the value of the spatial blocks (positive = joint better). The variable-capacity ceiling in regime R2 uses bounds $(x_{\min}, x_{\max}) = (50, 75)$ MW, calibrated to a loosely-binding regime on *training* data only (bind fraction ≈ 0.28 –0.37). Results in Table ??.

Case	ε^* (CV)	gap range (% of CVaR)	detectable	verdict
<code>us_west</code>	1 (8/9)	−0.005 ... +0.043	1/9 (trivial, $+3 \times 10^{-6}\%$)	null
<code>taskc</code>	1 (9/9)	−0.022 ... +0.038	4/9, <i>signs flip</i> across α	null (artifacts)
<code>us_hetero</code>	1 (9/9)	−0.233 ... +0.062	5/9, material ones <i>negative</i>	null (joint <i>hurts</i>)

Table 2: Spatial gap shuf – joint in CVaR_{0.95}, 9 cells per case (3 regimes $\times$ 3 risk levels α). Every gap is a small fraction of one percent. CV selects $\varepsilon^* = 1$ throughout, so the DRO is genuinely engaged — this is not a “model switched itself off” null. In `us_hetero` the only detectable cells are *negative*: estimating cross-region covariance that carries no exploitable signal merely adds estimation noise.

4 Reliability: robustness battery, multiple testing, temporal stability

Each case was re-run with seasonal-residual, AR(1)-residual, and Ledoit–Wolf-shrinkage covariance estimators. The largest $|\text{gap}|$ under *any* estimator is 0.07% (**us_west**), 0.18% (**taskc**), 0.36% (**us_hetero**) of CVaR — all economically negligible. No cell shows a *material* beneficial effect that is sign-stable across the seasonal *and* AR(1) estimators; the cells that flip positive disagree by orders of magnitude across estimators (e.g. **taskc** R3/ α 0.30: seasonal -0.033% vs. AR(1) $+0.179\%$), the signature of ε -selection noise.

Multiple testing. Testing 144 non-ablation cells, some bootstrap CIs exclude zero by chance. Applying a Benjamini–Hochberg correction at $q = 0.05$ over all cells (bootstrap p -values), the largest *positive* gap that survives is 0.179% — precisely the AR(1) cell already rejected by the agreement rule. Nothing material survives both criteria.

Temporal stability. A walk-forward refit (train 2021–2023, test 2024) reproduces the null: max $|\text{gap}|$ of 0.028%, 0.036%, 0.217% across the three cases. The result is not a 2025 artifact.

5 Mean-ablation: the causal mechanism

The preceding sections establish *that* the covariance adds no value; this experiment establishes *why*. We neutralize the mean field $\bar{p}$ used for *scheduling* while keeping *evaluation* on the real carbon panel. Two levels: LEVEL equalizes each region’s time-average to the global mean (keeping the hour-of-day shape); FLAT sets a constant mean everywhere, so $\langle \bar{p}, x \rangle$ is constant under fixed demand and the schedule is driven *purely* by the $\varepsilon \|L^\top x\|_2$ covariance penalty — a *covariance-only world*. Results in Table ??.

Case	baseline	LEVEL	FLAT (covariance-only)
us_west	~ 0 , none det.	~ 0 , none	$\varepsilon^* \rightarrow 0$; $-0.04 \dots +0.004$
taskc	~ 0 , sign-flips	~ 0	+0.14 $\dots$ +0.41 , all 9 det.
us_hetero	negative	negative	+0.39 $\dots$ +1.46 , all 9 det.

Table 3: Mean-ablation. Equalizing the *level* changes nothing — the diurnal mean shape still dominates. In the *covariance-only* world the joint covariance produces a material, fully-detectable advantage over the shuffled one, and CV no longer collapses to a single ε^* . The spatial covariance therefore *does* carry genuine, exploitable information — it is simply *swamped* by the mean in the real problem. Identical gaps on a $2\times$ denser ε grid confirm these are not grid artifacts.

This converts the null from an assertion into a demonstrated *ordering*: *the mean field dominates the schedule so completely that a real and exploitable second-moment signal contributes negligibly*. The cost confirms it is not free lunch: the FLAT schedule, by discarding the real mean, *worsens* absolute CVaR by +1.9–4.1% (**us_west**) and +1.6–3.8% (**us_hetero**) — the covariance-only gain ($\leq 1.46\%$) never repays it. **us_west** is the instructive exception ($\varepsilon^* \rightarrow 0$, gap ~ 0): its strong *common-mode* positive correlation offers little even in isolation, because diversification needs heterogeneity, not co-movement.

6 Tail dependence: the structure the covariance cannot see

Why is even the exploitable-in-isolation covariance the *wrong* object? Because the genuine dependence lives in the tails and is non-elliptical. We estimate the empirical upper/lower tail-dependence coefficients χ_U (both regions *dirty* together — the CVaR-relevant co-movement) and χ_L (both *clean* together), and compare χ_U against the Gaussian-copula value at the matched correlation. A Gaussian copula — anything a covariance ball can represent — has *zero* asymptotic tail dependence, so an empirical excess above it would be structure the DRO misses.

Case	Pair	ρ	χ_U^{emp}	χ_U^{Gauss}	excess	χ_L^{emp}
us_west	CISO LDWP	0.72	0.25	0.41	−0.16	0.48
us_west	BANC LDWP	0.65	0.17	0.35	−0.18	0.40
taskc	CA-ON NYIS	0.73	0.38	0.42	−0.04	0.31
us_hetero	ERCO BPAT	0.00	0.04	0.05	−0.02	0.02

Table 4: Residual tail dependence at $q = 0.95$. Two findings: (i) the empirical χ_U sits *at or below* the Gaussian benchmark ($\text{excess} \leq 0$) — in the dirtiest hours regions *de-couple* (upper-tail independence), so there is nothing in the dirty tail for a richer model to exploit; (ii) $\chi_L > \chi_U$ — regions go *clean* together more than *dirty* together. A Gaussian/elliptical copula forces $\chi_L = \chi_U$, so this radial asymmetry is non-elliptical structure a Mahalanobis–Wasserstein ball cannot encode.

7 Synthesis: why this is the expected outcome

1. **The mean field dominates** (proven in §??). The schedule is set by $\bar{\rho}$, shared identically by the joint and shuffled models; the covariance enters only as a second-order penalty.
2. **Positive correlation is common, non-diversifiable risk.** Diversification value needs regions that hedge each other; real grids co-move, and the genuinely clean escape (Ontario; hydro BPAT) is already identified by its low *mean*.
3. **Even zero correlation does not help** (us_hetero): with no cross-structure, fitting the joint covariance only adds estimation noise.
4. **The exploitable structure is non-elliptical** (§??) — invisible to a covariance/Wasserstein ambiguity set by construction.

8 Practical recommendation

For carbon-aware data-center scheduling under a covariance-based robust model, **a per-region marginal scheduler captures essentially all the value**. Modelling cross-region carbon dependence adds estimation burden and, in the heterogeneous case, measurable noise, for no robust benefit. Spatial dependence modelling should be reserved for (i) decision structures that can *act* on it (inter-region workload transfer) and (ii) dependence models that can represent its actual non-elliptical shape — both Phase 2 directions.

Phase 1 spatial-DRO summary: covariance adds no value (A); the tails are non-elliptical, not covariance-shaped (B, C)

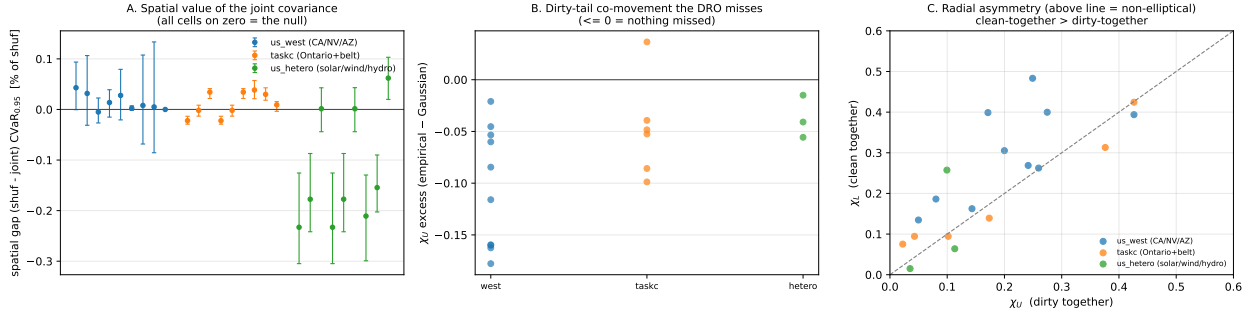


Figure 1: Phase 1 on one page. **(A)** spatial gap $\pm$ bootstrap CI for every case/regime/ α cell, clustered on zero — the null. **(B)** upper-tail dependence excess (empirical — Gaussian) ≤ 0 — no dirty-tail structure is missed. **(C)** χ_L vs. χ_U : points above the diagonal (clean-together > dirty-together) mark the non-elliptical asymmetry. Regenerate with `python -m scripts.plot_case_summary`.

9 Conclusion and Phase 2 outlook

Phase 1 is a *specification result*: the second moment of spatial carbon dependence is both *dominated* (by the mean field) and the *wrong object* (the dependence is non-elliptical — upper-tail-independent and radially asymmetric). This is not a dead end but a precise mandate for Phase 2: model the **copula**, not the covariance. A vine-copula ambiguity set can represent asymmetric tail dependence ($\chi_L \neq \chi_U$, Clayton-type clean-together coupling) that the Mahalanobis–Wasserstein ground metric cannot express; in parallel, an inter-region *transfer* channel would give the decision a spatial degree of freedom on which the joint distribution could finally act. Phase 1 has both falsified the simple spatial-covariance hypothesis and identified, with evidence, exactly where the remaining value must lie.

Reproducibility. All results derive from `scripts/run_case_experiment.py` (region sets `us_west`, `taskc`, `us_hetero`; flags `--residualize`, `--shrinkage`, `--ablate-mean`, `--test-year`, `--eps-grid`), `scripts/calibrate_capacity.py`, `scripts/bh_correction.py`, and the diagnostic plotters. Summary tables for every number cited here are archived in `docs/results_snapshots/`.